

Mini Game Project – Tic Tac Toe

Introduction

The Tic Tac Toe Game is a console-based mini project developed using C++. The project is designed to demonstrate fundamental programming concepts such as loops, arrays, functions, and conditional statements through an interactive two-player game. The game allows players to take turns marking spaces on a 3×3 board until one player wins or the game ends in a draw.

Objectives

- To understand the implementation of arrays and loops.
 - To apply conditional statements for game logic.
 - To develop problem-solving and logical thinking skills.
 - To create an interactive console application in C++.
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Features

- Dynamic 3×3 game board display.
 - Two-player gameplay system.
 - Input validation for correct moves.
 - Automatic win and draw detection.
 - Replay option after game completion.
 - Simple and user-friendly console interface.
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Technologies Used

- Programming Language: C++
 - Platform: Console/Terminal
 - Concepts Used:
 - Arrays
 - Loops
 - Functions
 - Conditional Statements
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Working of the Project

1. The game board is initialized with positions from 1 to 9.
 2. Player X starts the game first.
 3. Players enter the position number to place their mark.
 4. The board updates dynamically after each move.
 5. The program checks for:
 - Winning condition
 - Draw condition
 6. If a player wins or the game ends in a draw, the result is displayed.
 7. Users can choose to replay the game.
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Sample Output

```
1 | 2 | 3
---|---|---
4 | 5 | 6
---|---|---
7 | 8 | 9
```

Player X, enter position: 5

```
1 | 2 | 3
---|---|---
4 | X | 6
---|---|---
7 | 8 | 9
```

Player O, enter position: 1

```
O | 2 | 3
---|---|---
4 | X | 6
---|---|---
7 | 8 | 9
```

Player X wins!

Expected Outcome

The project successfully creates an interactive Tic Tac Toe game using C++. It demonstrates the effective implementation of game logic, dynamic board updates, player interaction, and win/loss detection in a console environment.

Conclusion

The Tic Tac Toe Mini Game Project helps beginners understand core C++ programming concepts in a practical and engaging way. It improves logical thinking, coding skills, and knowledge of game development basics using simple console programming.